

Module 04: Cognition in Crochet Circles

Learning Objectives

1. Explain how reading a crochet pattern engages a **generative model** that predicts fabric structure before a single stitch is made.
2. Describe **mental stitch counting** as a form of internal simulation — running the generative model forward in time without acting.
3. Connect the experience of "**thinking with your hands**" to embodied cognition in the Active Inference framework.

Introduction

Crochet is not mindless work. Even the simplest dishcloth requires a surprising amount of cognitive activity: reading a pattern, holding a stitch count in memory, predicting how the fabric will behave, imagining the finished object. In Active Inference, **cognition** is the process by which an agent uses its generative model to simulate, predict, and plan — to think about the world without necessarily acting on it. When you sit with a pattern and mentally walk through the instructions, when you count stitches in your head, when you predict that a certain yarn will drape beautifully in a shawl, you are performing cognitive operations that Active Inference describes with mathematical precision.

This module explores the thinking that happens alongside (and sometimes instead of) stitching. We will look at pattern reading as model engagement, mental simulation as internal prediction, and the fascinating phenomenon of "thinking with your hands" — where cognition is distributed between brain and body.

Key Concepts

1. Pattern Reading as Generative Model Engagement

When you open a crochet pattern, you encounter a compressed description of a future object: "Ch 4, join with sl st to form ring. Rnd 1: Ch 3 (counts as dc), 11 dc in ring, join with sl st to

top of ch-3. (12 dc)." Reading this engages your generative model — the internal representation that allows you to predict what the fabric will look like at each stage.

In Active Inference, a generative model specifies how hidden states (the structure of the fabric, the arrangement of stitches) generate observations (what you see, feel, and count). When you read a pattern, you are essentially **loading parameters** into your generative model. Each instruction — "dc in next st," "skip 2 sts," "3 dc in corner sp" — adds a constraint that shapes your predictions.

Experienced crocheters read patterns like musicians read sheet music: they can "hear" the piece before playing it. They can look at "ch 3, skip 2, dc in next st, *ch 1, skip 1, dc in next st, repeat*" and immediately visualize a filet mesh pattern with its characteristic grid of open and filled squares. This ability to translate symbolic notation into predicted fabric structure is a cognitive feat — the generative model translating compact instructions into rich, multi-dimensional predictions about texture, shape, drape, and appearance.

Beginners lack this translation ability. They read each instruction as an isolated action: "chain three... now skip two... now double crochet." They cannot yet predict what the accumulation of these actions will produce. The generative model is too sparse to generate look-ahead predictions. As experience grows, the model deepens, and the crocheter can increasingly "see" the finished product in the pattern instructions — a hallmark of cognitive expertise.

2. Mental Stitch Counting as Internal Simulation

Internal simulation in Active Inference is the process of running the generative model forward in time without actually performing actions or receiving new sensory input. It is thinking, imagining, planning — using the model to explore possibilities.

Mental stitch counting is a clear example. When a pattern says "sc in each of next 15 sts," you do not need to physically count 15 stitches before starting — you hold "15" in your working memory and count down (or up) as you stitch. Each stitch decrements an internal counter. This counter is a mental simulation running in parallel with the physical action: the model predicts that after 15 motor actions (insert, yarn over, pull through), the row should be complete.

More complex simulation occurs when crocheters plan ahead. You are working a granny square and need to decide how many rounds to make before it is the right size. You imagine

adding another round — your generative model predicts how much larger the square will become, based on your knowledge of how granny rounds grow (each round adds about 1-2 inches, depending on yarn weight and hook size). You simulate the outcome without actually crocheting it. If the simulation predicts the right size, you proceed. If not, you adjust.

This kind of simulation is what Active Inference calls **counterfactual reasoning**: evaluating outcomes of actions you have not yet taken. Every time a crocheter thinks "If I add one more repeat of the pattern, will it be long enough?" they are running their generative model in simulation mode, generating predicted observations (length, width, shape) and comparing them to their goals.

3. Thinking with Your Hands: Embodied Cognition

One of the most striking features of crochet cognition is how much of it happens in the hands rather than the head. Experienced crocheters report that their hands "know" what to do — they can hold a conversation, watch television, or listen to an audiobook while crocheting complex patterns. The hands seem to think independently.

In Active Inference, this is not mysterious — it is a natural consequence of how generative models can be distributed across the body. The brain is not the only location of the generative model; the hands, through years of practice, have developed their own local models of stitch mechanics. The hands predict what each stitch should feel like, execute the motor sequence automatically, and detect errors through tactile prediction errors — all without requiring conscious, top-down cognitive processing.

This is **embodied cognition** in the Active Inference framework. The generative model is not a single, monolithic structure sitting in the brain. It is a hierarchy of models distributed across levels: high-level models in the brain handle pattern-level predictions (which stitch comes next, how the pattern repeats), while low-level models in the motor system handle stitch-level predictions (how much force to apply, where to insert the hook, how to wrap the yarn). When these levels are well-calibrated, crochet feels effortless — the "thinking" is distributed.

The moment something unexpected happens — an unfamiliar stitch, a pattern error, a yarn tangle — cognition snaps back to conscious processing. The hands stop, the crocheter looks down, and top-down cognitive resources are deployed. This shift from automatic to controlled

processing is precisely what Active Inference predicts: when low-level models generate prediction errors they cannot resolve, the errors propagate upward to higher-level models, demanding conscious attention.

Applications

In crochet, we see Cognition manifest in:

- **Predicting How Fabric Will Drape:** Before finishing a garment, experienced crocheters can predict whether it will hang well or bunch up, whether it will stretch out or hold its shape. This prediction is based on their generative model's knowledge of how different stitch patterns, yarn fibers, and hook sizes interact to produce specific fabric properties. A crocheter who chooses a lace stitch for a summer top is cognitively simulating drape and breathability before making a single stitch.
- **Mental Pattern Modification:** Many crocheters modify patterns as they work — adjusting length, changing stitch counts, substituting stitches. This requires real-time cognitive simulation: "If I add 6 more stitches to the foundation chain, how will that change the width? Will the pattern repeat still work evenly?" The crocheter runs their generative model in simulation mode, checks whether the modification is consistent, and then executes (or rejects) the change. This is sophisticated counterfactual reasoning grounded in embodied knowledge.

Conclusion

Cognition in crochet is richer than it appears. Pattern reading loads your generative model with predictions. Mental stitch counting runs that model as an internal simulation. And the remarkable phenomenon of "thinking with your hands" reveals that the generative model is not confined to the brain — it is distributed across the entire body-hook-yarn system. In the next module, we move from thinking to doing: how each stitch is an action that changes the fabric and the world.